ASSIGNMENT - 01

Introduction

The internet has become a universal medium and people now often choose to rent property online. But many landlords continue to manage properties using outdated practices like paper-based systems and stand-alone software applications. This results in delays, communication gaps, and inefficiencies in house renting. This contributes to a lack of seamless information flow, and renters can face complexities.

A web-based house rental system can help in this regard. An online system automates every aspect of renting, such as property search, reservation and payment. It offers an integrated platform enabling efficient communication and coordination between tenants and owners. Also, online systems contribute to security through secure payment options.

It offers a dynamic digital interface, eliminating traditional inconvenient methods, for timely processing and better responsiveness. Its intuitive interface makes it easier to complete tasks like registration and renewing leases. Additionally, the system is scalable so that it can grow to meet demand. Through its well-integrated design, the system guarantees secure and efficient data processing, helping to gain users' trust in the long run.

Problem Statement

The existing rental system is inefficient. It often includes in-person meetings, multiple phone calls and the need for brokers, even though they might not be needed. Potential renters must visit repeatedly to see multiple properties, resulting in time and energy being wasted. Further, paper-based documents are generating substantial paperwork and delays.

Lastly, there is a lack of information on property availability, pricing and rental agreements. Existing platforms do not always provide timely information, leading to inaccurate listings and information. Rentals are sometimes booked through complex and error-prone processes, with the result of rental transactions failing or rooms being over-booked. These factors lead to customer dissatisfaction and loss of confidence.

To overcome these problems, we need to develop a system that consolidates all the rental processes. This system should facilitate online property listings with accurate and timely details of rates, conditions and availability. It can facilitate easy booking management without multiple back-and-forth messages and provide online payment security.

Additionally, it should have an admin interface to track property listings, enforce the booking schedule, and keep track of tenants. Through this integration, the system can streamline processes, enhance transparency, and provide a reliable and convenient rental experience.

Key Challenges

1. Time-Consuming Property Search
Searching for a rental property that matches a renter's requirements can be a frustrating task. Listings are not properly categorised and unavailable properties are not properly matched with user preferences, making the search inefficient and unpredictable.

2. Uncertain Information and Transparency
Users often have little information about aspects like lease terms, prices and availability. Key terms may be hidden until the later stages, causing confusion and frustration. Also, price variations and inaccurate or outdated listings can deceive users and erode the platform's credibility.

3. Inefficient and Fragmented Processes
Manual processes and fragmented technology-based systems can slow down and make processes inefficient. Such disconnected processes, in turn, lead to delays in transactions and frustration from both customers and service providers.

4. Lack of Integration and Accessibility
Legacy systems are often not integrated with contemporary web and mobile systems. This makes system access less convenient for users, and hampers its scalability and potential for expansion.

How This System Fixes the Issue

The system offers a comprehensive, online platform that simplifies the house rental process by digitising property search, booking and payment. The automation of core processes enhances efficiency, allowing users to easily search for properties, make reservations and conduct transactions.
The system provides greater transparency by providing timely information on property availability, rental rates and terms. This reduces doubts and confusion. The system also eliminates paper-based operations, enabling quicker and more efficient data management through a digitized system.
The system's intuitive design makes it easy for end-users and administrators to interact with it, with straightforward navigation for viewing real estate properties, reserving properties, and maintaining records. An intuitive process flow helps avoid confusion and boosts user experience.
Overall, the centralisation of all rental processes in one digital package guarantees a seamless, efficient and reliable user experience.

Key features and solutions

1. Powerful Search and Filter The system offers a straightforward search and filter system, enabling users to easily locate rental properties. Search results can be ordered and filtered by various criteria such as price, location, property type (e.g. apartments, cabins), length of stay and desired dates.The filter system helps to streamline the search process, presenting only relevant properties without distractions. What's more, the website has a robust back-end structure that provides real-time availability updates, reducing confusion and the risk of double bookings. This leads to a more efficient and seamless search process.

2. Clear Price and Rental Terms The system provides full transparency by prominently displaying prices and rental policies for properties. Rental periods, rules, security deposits, and cancellation and refund policies are readily accessible to users before finalising their reservations.This upfront presentation of information prevents hidden fees and surprises, allowing users to make well-informed choices. This boosts user confidence and minimises the risk of disputes during rentals..

3. Efficient Booking Process It offers a convenient and efficient booking process by reducing input duplication. User data is collected and stored securely and can be used for subsequent bookings without having to re-enter it. This enhances user experience and speeds up the booking process.The system provides a structured overview of all relevant booking information, such as property details, rental price, length of stay and dates. It also clearly shows payment information.Through the integration of all the necessary details on a single screen and the ability to confirm swiftly, the system provides users with a seamless, efficient, and convenient booking process.

4. Secure Payment Processing The system ensures that all financial transactions are conducted through trusted and secure payment gateways, maintaining high standards of data protection. Sensitive user and payment information is encrypted and handled securely to prevent unauthorized access.The platform supports multiple payment methods, including credit cards, debit cards, and digital wallets, providing flexibility and convenience to users. Regardless of the chosen method, each transaction is processed efficiently and securely.By offering diverse and reliable payment options, the system ensures a seamless and safe payment experience for all users

5. Seamless Backend Integration The platform facilitates the integration of all backend components, allowing real-time updating of property availability, bookings and user information. Data is kept in sync, ensuring data integrity and consistency.Any updates in one part, such as bookings or payments, are automatically reflected system-wide, eliminating inconsistencies and lag times. Update processes and patches also increase reliability by automating processes and preventing human error. This streamlines operations and facilitates seamless use of the system.

6. Accessibility and Responsiveness
The system is responsive and accessible on multiple devices such as mobile phones, tablets, laptops and desktops. It has a flexible user interface that responds to different screen sizes and orientations, providing a consistent user experience.The system features a simple and straightforward navigation system for easy access to functionality. The system is user-friendly and accessible with features that accommodate users of different abilities. The system provides a seamless, user-friendly experience for all types of users on all devices.

Benefits to Users

- **End-to-End Online Booking** This allows users to book and manage all aspects of property rental - search, booking, payment - in one platform. This eliminates the need for agents and eliminates the reliance on paper documentation, allowing for a quicker and more efficient process.

- **Clarity and Confidence** Users can view all necessary information, such as price, property specifications, lease terms and availability. This enables users to make informed choices without any doubts or hidden surprises post confirmation.

- **Quick and Effective Search** Regular search and filtering options allow users to easily find the right property for their needs. This saves time and ensures that the search process is efficient and effective, enhancing the user's experience.

- **Enhanced Security and Trust** It provides secure login and payment transactions to enhance security in handling user information and payments. This guarantees privacy and instills confidence by offering a secure platform for users.

- **Flexibility and User Control** Users can select properties according to their requirements, preferences and stay length. The system offers flexibility, enabling users to change their selections as needs arise, providing better user satisfaction and control over the rental experience.

Benefits to Businesses

- **Time Saving and Efficiency** The system automates crucial operations such as property management, booking and record keeping. This eliminates the need for manual participation, reduces repetitive processes and speeds up operations. It also reduces the risk of errors, enhancing accuracy.

- **Enhanced Customer Loyalty** Convenient and efficient service increases customer satisfaction. Being able to easily search for and reserve properties instills confidence in the user. This, in turn, leads to increased customer loyalty and retention.

- **Real-Time Data Management** Listing details, availability, and bookings are automatically updated in real-time, ensuring that the data is always current and accurate without the need for continuous monitoring. This enables businesses to stay consistent and avoid problems like stale listings or double bookings.

- **Scalability and Growth Support** It can manage a growing user base and property listings without slowing down. As user and listing volumes increase, the system remains responsive, enabling businesses to grow without sacrificing performance.

- **Consistent Performance Under Load** It continues to perform well and respond promptly even under heavy load. This guarantees continuous operation and a consistent user (and admin) experience, even under high loads.

Fixing what slows down old-style home rentals, this online system makes renting quicker, safer, simpler. Built to grow smoothly, it helps renters and landlords alike. Now, managing property leases feels fresh - clearer, smoother, sharper than before.

ASSIGNMENT - 02

Software Requirement Specifications (SRS) Document

Online House Rental Management System

1.Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document aims to specify the functional and non-functional requirements of the Online House Rental Management System. This document will be used by developers, designers, testers and stakeholders in the development and maintenance of the system.This is a dynamic system that allows renters to search, view, and book rental properties online. It enables property owners and administrators to handle property listings, availability, bookings, tenant information, and payments in one system. In general, the system aims to enhance the rental process, efficiency, and organisation for users.

1.2 Intended Audience

This Document is intended for:

- Project stakeholders
- Developers and testers
- System designers
- Faculty evaluators
- Future maintenance teams
- Landlords and property managers (for understanding system scope)

1.3 Intended Use

The system will be used to:

- Search and filter rental properties
- View detailed property information
- Reserve and book properties
- Enter renter information and required documents
- Make secure online payments
- Manage rental listings, bookings, and availability (admin/landlord side)

1.4 Product Scope

The Online House Rental Management System addresses common problems of manual rental processes such as:

- Delayed updates
- Unclear property availability
- Hidden rental rules
- Double bookings
- Slow communication
- Paper-based documentation

The system provides a complete digital workflow: **Listing → Searching → Booking → Payment → Rental Record Maintenance**

1.5 Definitions, Acronyms, and Abbreviations

- **SRS** – Software Requirements Specification
- **UI** – User Interface
- **Admin** – User responsible for managing listings, bookings, and tenant records
- **Tenant/Renter** – User who rents a property
- **Landlord** – Property owner who lists properties
- **Booking** – Reservation of a property for a specific time period
- **WCAG** – Web Content Accessibility Guidelines

1.6 References

The following documents and resources were referred to and inspired the design and requirements of this system:

- WCAG 2.1 Level AA – Web Content Accessibility Guidelines (W3C)
- Stripe / Razorpay / PayPal API documentation (for payment workflow reference)
- MDN Web Docs – Web application design and security practices

2.Overall Description

2.1 Product Perspective

The system is a **web-based rental management application** that replaces traditional renting methods such as phone calls, brokers, and paper forms with a streamlined digital solution.

System components include:

- **Frontend:** React with Vite and TypeScript for building responsive user and admin interfaces
- **Backend:** Node.js with Express framework providing RESTful APIs for business logic and authentication
- **Database:** MongoDB for storing users, properties, bookings, availability schedules, and payment records

2.2 Product Features

Key features of the system include:

- **Property Browsing** – View properties with images, rent, location, type, and availability
- **Search and Filters** – Filter properties by location, rent range, house type, rooms, furnishing, and availability
- **Property Details View** – Display amenities, rules, deposits, and policies
- **Booking Process** – Reserve properties by selecting dates and entering renter details
- **Secure Payments** – Online payment processing with confirmation and receipts
- **Admin/Landlord Management** – Manage listings, bookings, tenants, and payments

2.3 User Characteristics

- **Renters:** Basic smartphone or PC users seeking fast and transparent booking
- **Landlords:** Users managing property listings and availability
- **Admin:** System manager responsible for verification, approvals, and conflict resolution

2.4 Constraints

- Must support modern browsers (Chrome, Firefox, Edge, Safari)
- Must comply with payment security standards
- Responsive design for mobile and desktop devices

2.5 Assumptions and Dependencies

- Users have internet access and valid payment methods
- Third-party payment gateway services are available
- Email/SMS services are available for notifications

3. System Features and Requirements

3.1 Functional Requirements (EARS Format)

3.1.1: User Registration and Login

- When a user wants to access the platform, the system shall allow registration using email or phone number.
- When valid credentials are entered, the system shall authenticate the user.
- If credentials are invalid, the system shall display an error message.

3.1.2: Browse Property Listings

- Where listings exist, the system shall display properties with rent, location, type, images, and availability.
- When a property is selected, the system shall display full property details.

3.1.3: Search Properties

- When a user enters a keyword, the system shall return matching property results.

3.1.4: Apply Filters

- When filters are applied, the system shall update the search results accordingly.
- Filters include rent range, property type, furnishing status, availability date, and number of rooms.

3.1.5: View Property Details

- The system shall display rent, deposit, availability, house rules, and cancellation policies.

3.1.6: Booking System

- When a renter confirms booking dates, the system shall reserve the property.
- If a property is already booked, the system shall prevent double booking.

3.1.7: Renter Information Entry

- The system shall collect renter details such as name, phone number, and required identification.

3.1.8: Payment Processing

- The system shall redirect users to a secure payment gateway.
- Upon successful payment, the system shall confirm the booking and generate a receipt.
- If payment fails, the system shall cancel the booking request.

3.1.9: Booking History

- The system shall allow users to view booking history and payment status.

3.1.10: Admin Management

- The system shall allow admins to add, edit, approve, or remove property listings.
- The system shall allow admins to manage bookings, availability, and tenant records.

3.2 Non-Functional Requirements

Performance

- Pages should load within 2 seconds under normal conditions.
- Search and filter responses should occur within 1 second.

Security

- Payment processing must comply with PCI-DSS standards.
- Passwords must be securely hashed.
- User sessions must expire after inactivity.

Usability

- The UI should be simple and intuitive.
- Forms should support validation and autofill.

Reliability

- Booking and payment data must not be lost.
- The system should handle high traffic reliably.

Compliance

- WCAG 2.1 AA accessibility compliance
- Data privacy and minimal data storage

4. Interface Requirements

4.1 User Interfaces

- **Home Screen:** Search bar, featured listings, filters
- **Property Details Screen:** Images, rent, deposit, rules, amenities, booking button
- **Booking Screen:** Renter details form, date summary, payment options
- **Admin Dashboard:** Listing management, booking records, tenant information

4.2 Hardware Interfaces

- Smartphones
- Tablets
- Laptops and desktops

4.3 Software Interfaces

- Payment gateway APIs (Stripe, Razorpay, PayPal, etc.)
- Email and SMS notification services
- Backend REST APIs

5. Other Requirements

5.1 Database Requirements

The database shall store:

- User accounts
- Property listings
- Booking records
- Payment transactions
- Availability schedules

5.2 Localization

- English language support initially
- Multilingual support can be added in future versions

5.3 Accessibility

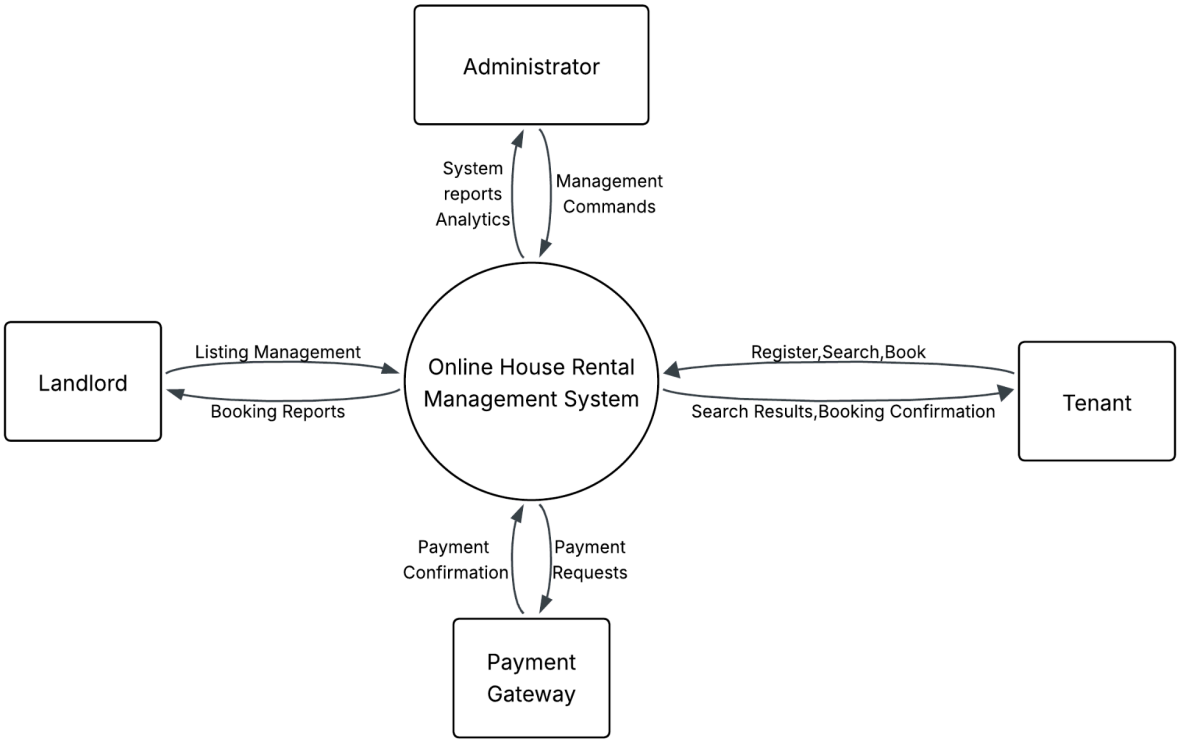
The system shall follow WCAG 2.1 AA guidelines to ensure inclusive access for all users. Key accessibility considerations include:

- Keyboard accessibility and logical navigation flow for users who cannot use a mouse
- Screen reader compatibility through proper semantic HTML, labels, and ARIA roles
- Adequate color contrast, readable typography, and responsive layouts to support users with visual impairments

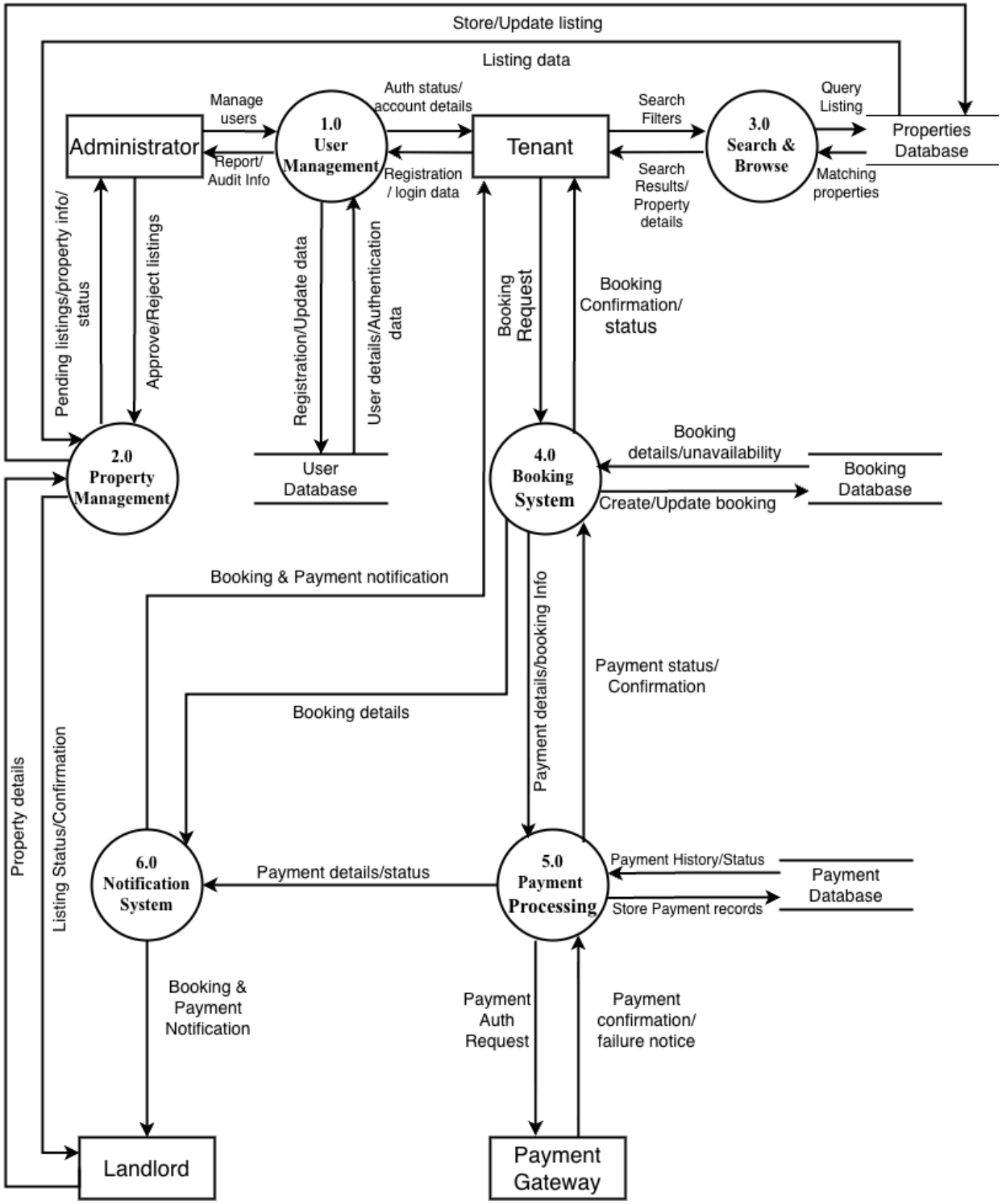
ASSIGNMENT - 03

ER diagram of Rental Management System

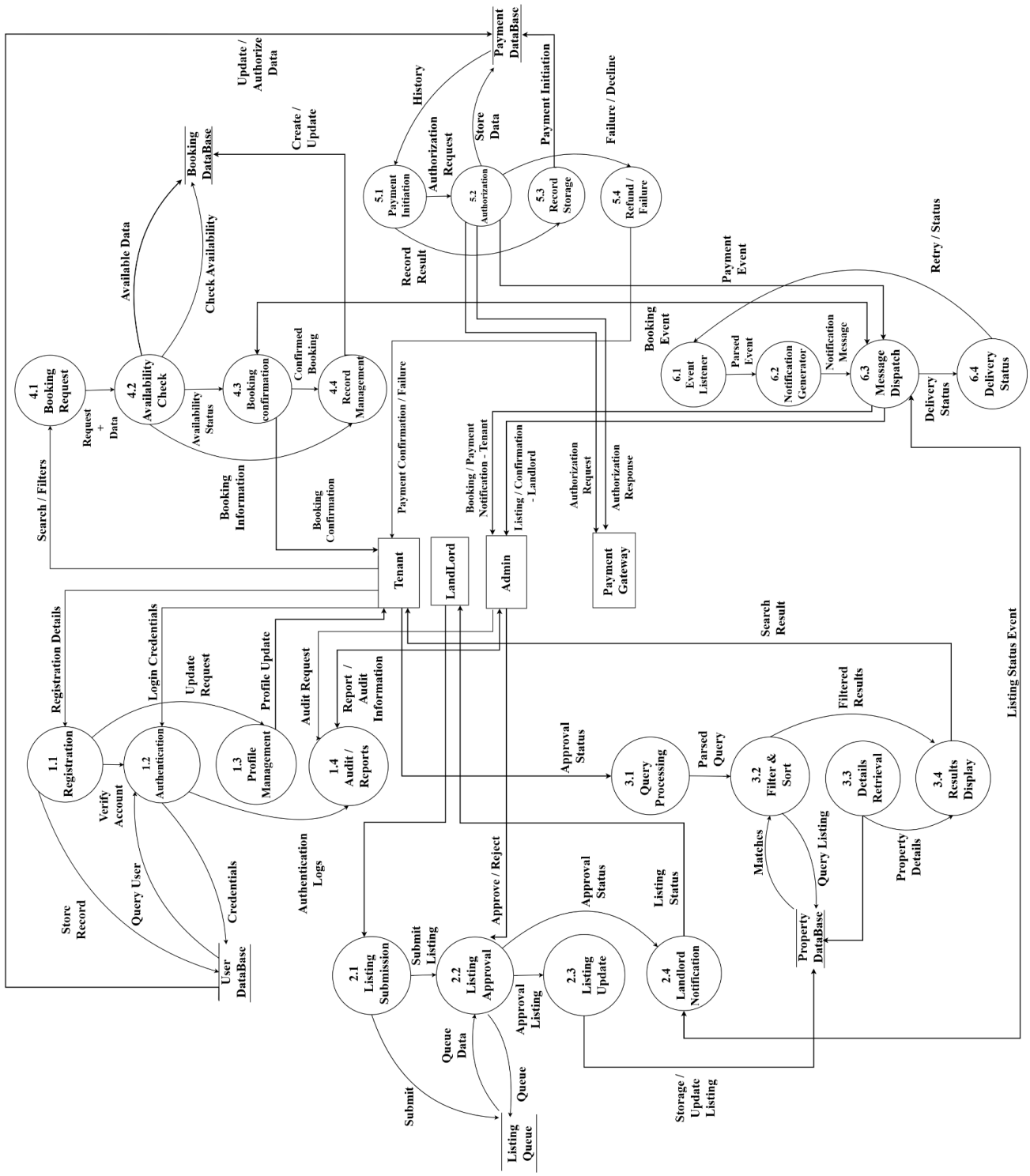
Level 0 Data Flow Diagram (DFD) of Rental Management System



Level 1 Data Flow Diagram (DFD) of Rental Management System

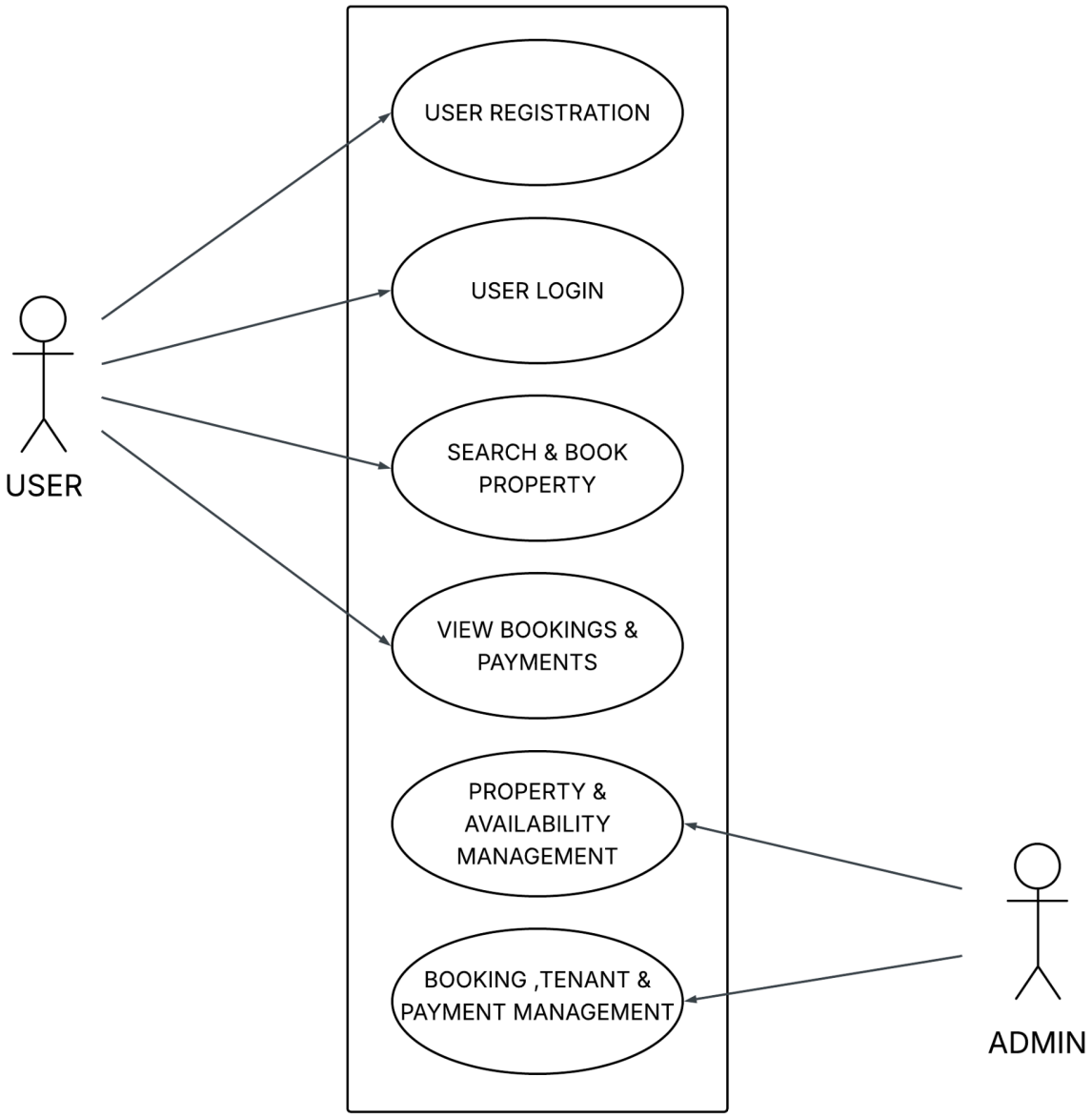


Level 2 Data Flow Diagram (DFD) of Rental Management System



ASSIGNMENT - 04

Use Case of Online Rental Management System



Description of Use Cases

1. User Registration

Precondition: User has not created an account yet.

Main flow:

- User opens the app and selects "Register."
- User completes his/her registration by filling out the registration form with name, email, mobile number and password.
- System validation of the input and duplicate email or mobile number.
- In case of no duplicate, then the system will create a unique User ID and validate the account.
- Registration is successful and the user is redirected to the login.

Error flow:

- In case a duplicate email or cell number is identified, the user is shown an error message and asked to re-enter with other details.

2. User Login

Precondition: User must have a registered account.

Main flow:

- User enters the app and presses on the option of "Login."
- User needs to key in his/her email or cell number and password.
- System checks the credentials against database.
- On credential validation, the system allows access and shows the dashboard.

Error flow:

- In case of incorrect credentials or absence of account the system sends an error message and will request user to re-attempt.

3. Search & Book Property

User search and book properties.

Main flow:

- Enter details => System will show results.
- Select property => Enter dates.
- System will check dates; if available, system will create booking and show cost.
- After that, user will be asked to pay; once payment is done, system will show booking confirmation.

Not available:

- Conflict detected; show "Property not available."
- Select different dates/property.

4. View Bookings & Payments

User will be able to view bookings/payments.

Main flow:

- Go to "My Bookings/Payments."
- The system will show records.
- View/download receipt.

No records:

- Show "No records found."
- Go to properties to search.

5. Property & Availability Management (Admin)

Admin will be able to manage properties/availability.

Main flow:

- Admin will login => Go to "Manage Listings."
- Add/edit property => System will validate/check duplicates.
- Property saved => Update availability; no conflicts detected.
- Update successful.

Errors/Conflicts:

- Duplicate property detected or conflict detected.
- Show error messages; update details.

6. Booking, Tenant & Payment Management (Admin)

Admin will be able to manage bookings/tenants/payments.

Main flow:

- Admin will view bookings/tenants/payments => Go to “Manage Bookings/Tenants/Payments. ”
- View bookings/tenants => Approve/cancel bookings; view tenant details/history.
- View transactions => Generate QR codes.
- Update records.

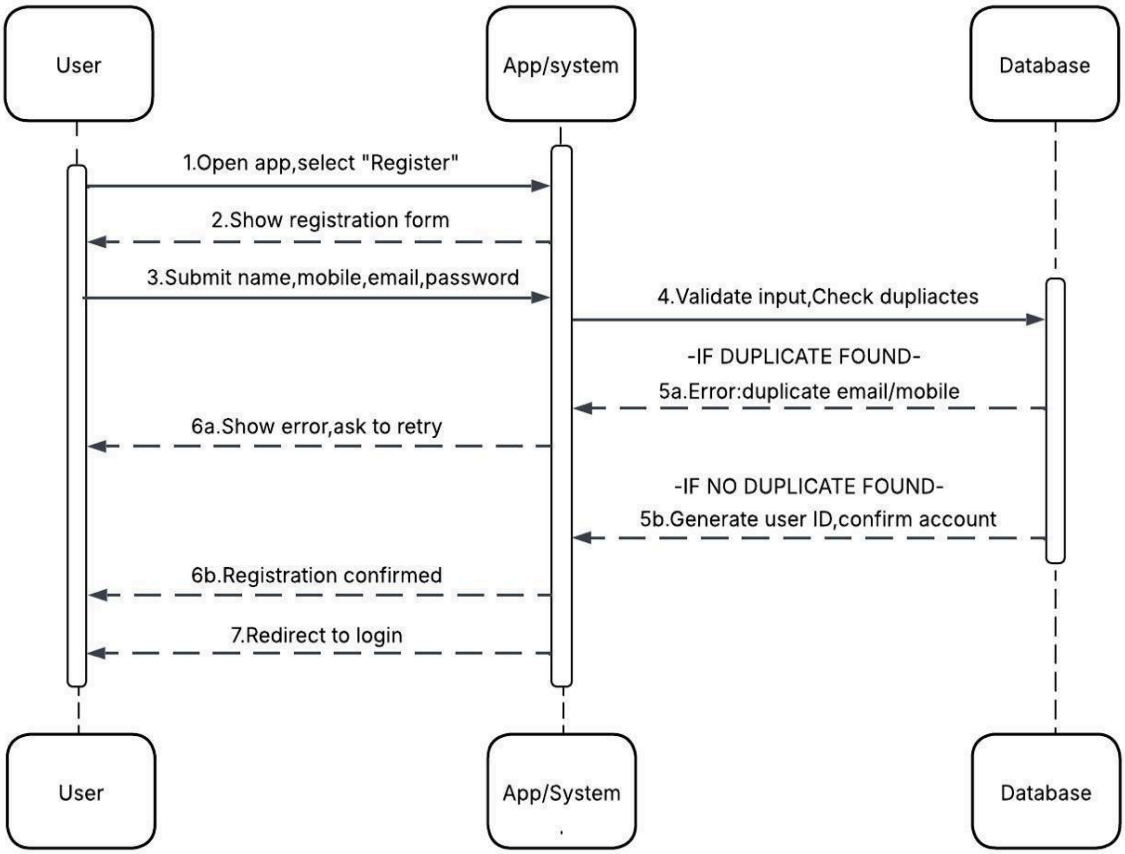
Restrictions/flags:

- Deleting tenants with existing bookings not allowed.
- Suspicious transactions detected; show warning messages.

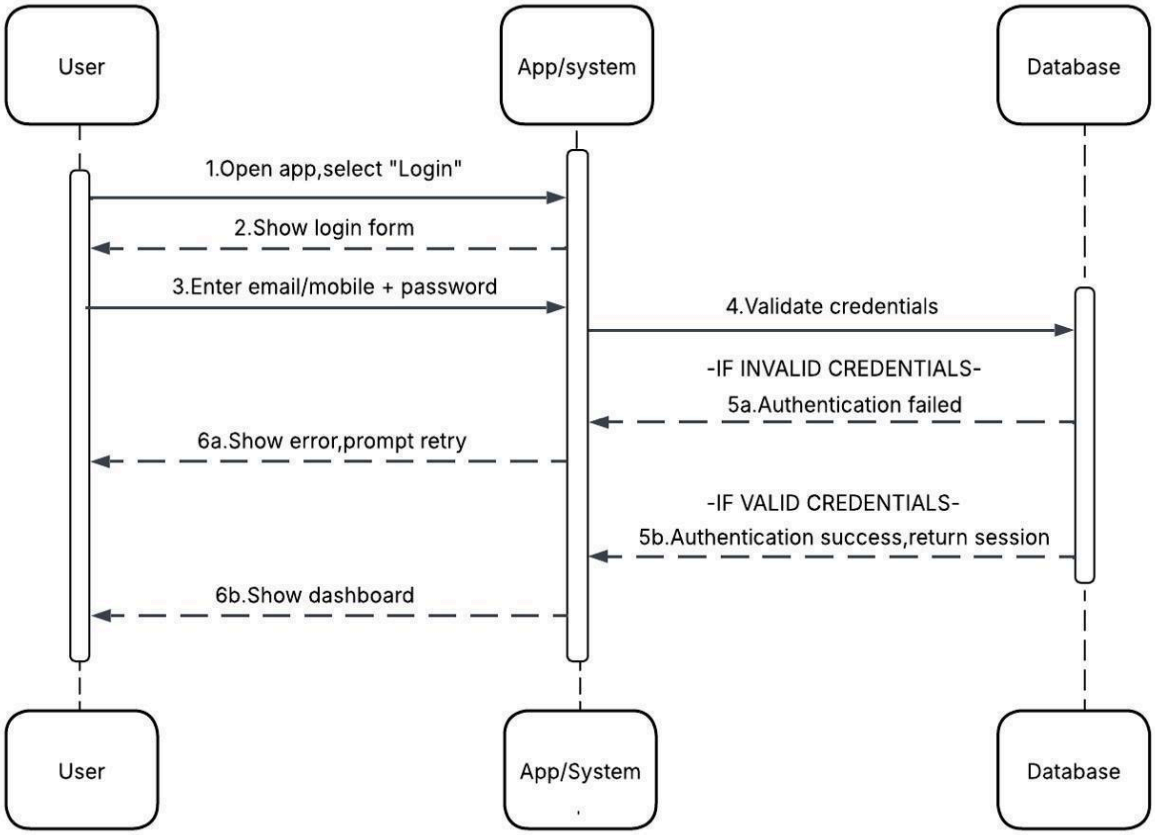
ASSIGNMENT - 05

Sequence Diagrams

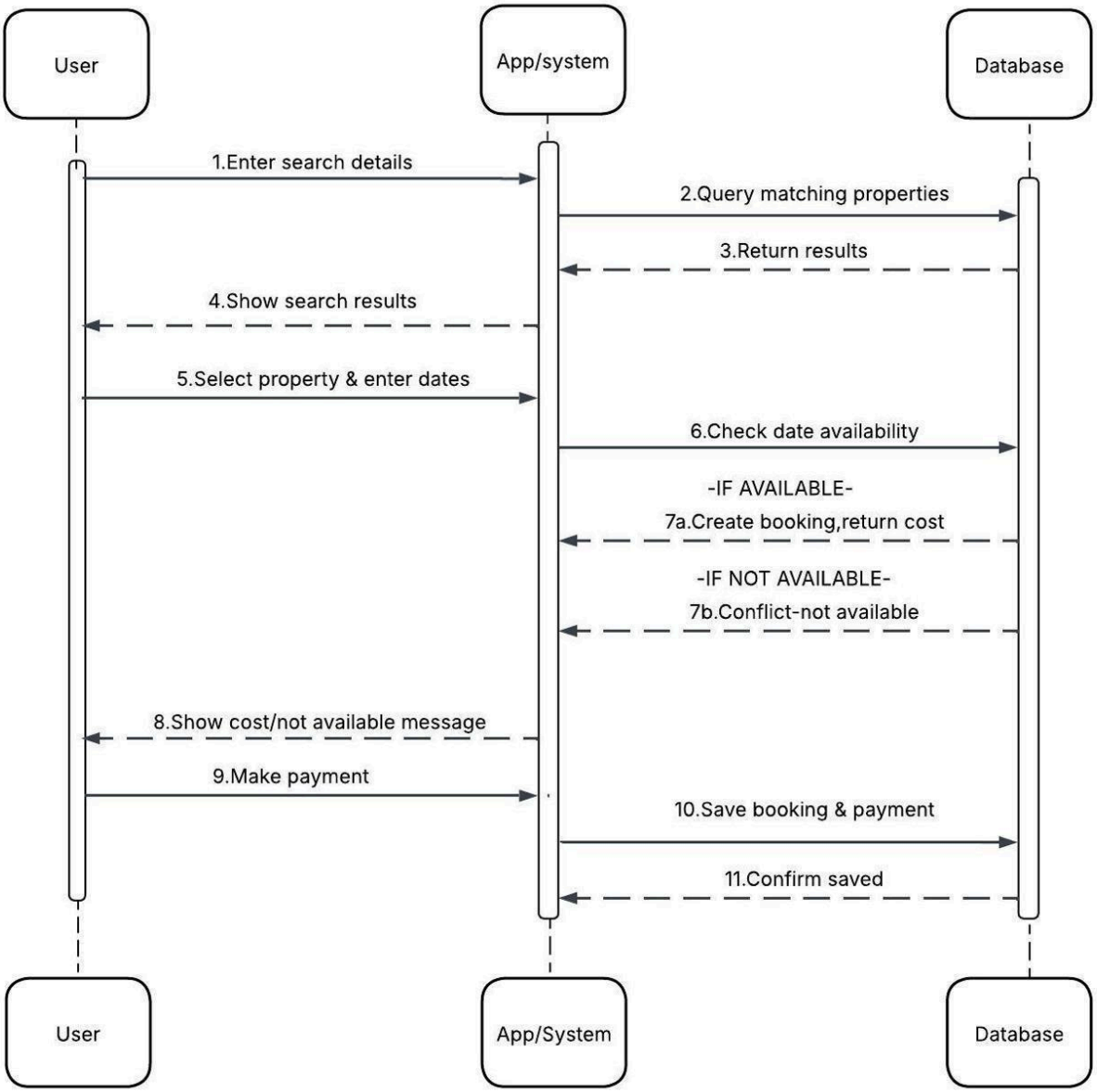
1. User Registration



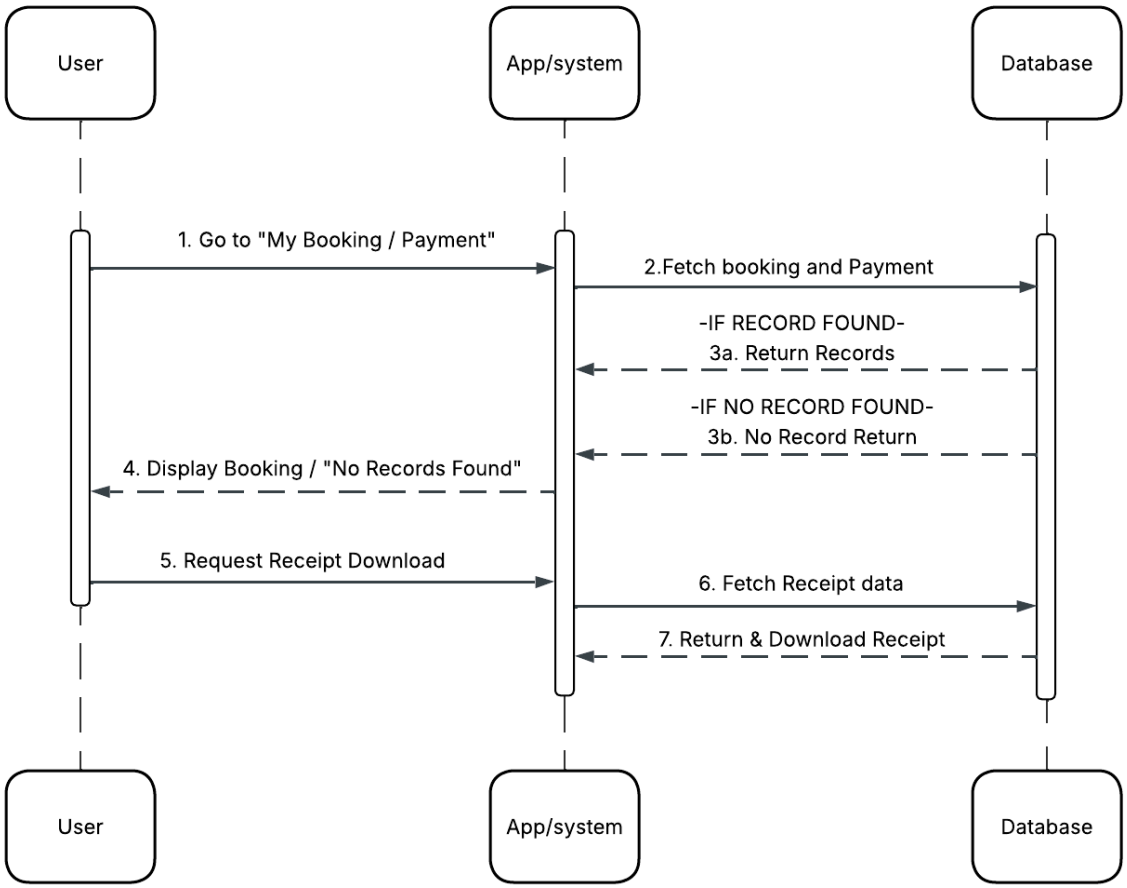
2. User Login


User
App/system
Database
1.Open app,select "Login"
2.Show login form
3.Enter email/mobile + password
4.Validate credentials
-IF INVALID CREDENTIALS-
5a.Authentication failed
6a.Show error,prompt retry
-IF VALID CREDENTIALS-
5b.Authentication success,return session
6b.Show dashboard
User
App/System
Database

3. Search & Book Property

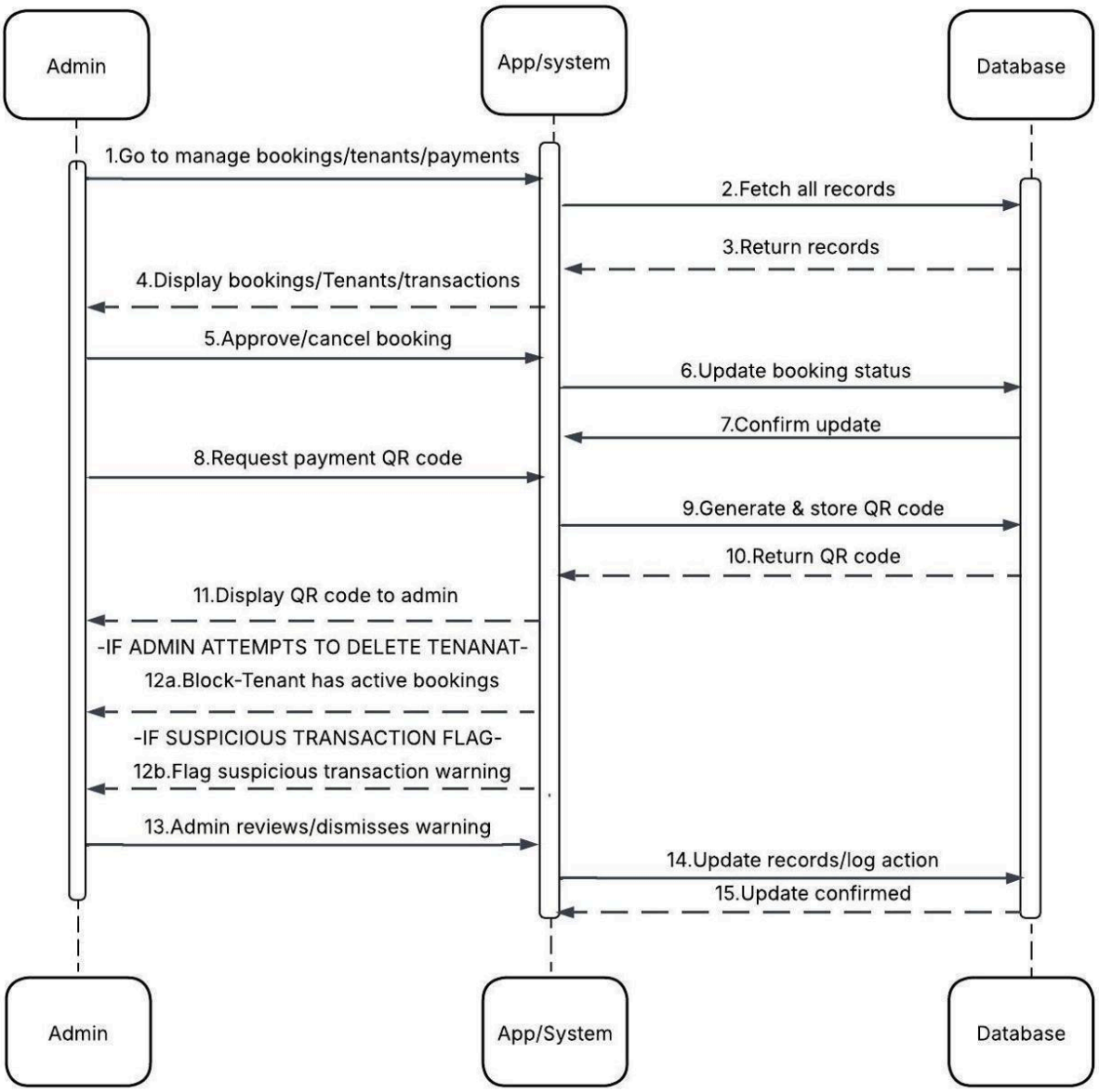


4. View Bookings & Payments



5. Property & Availability Management (Admin)

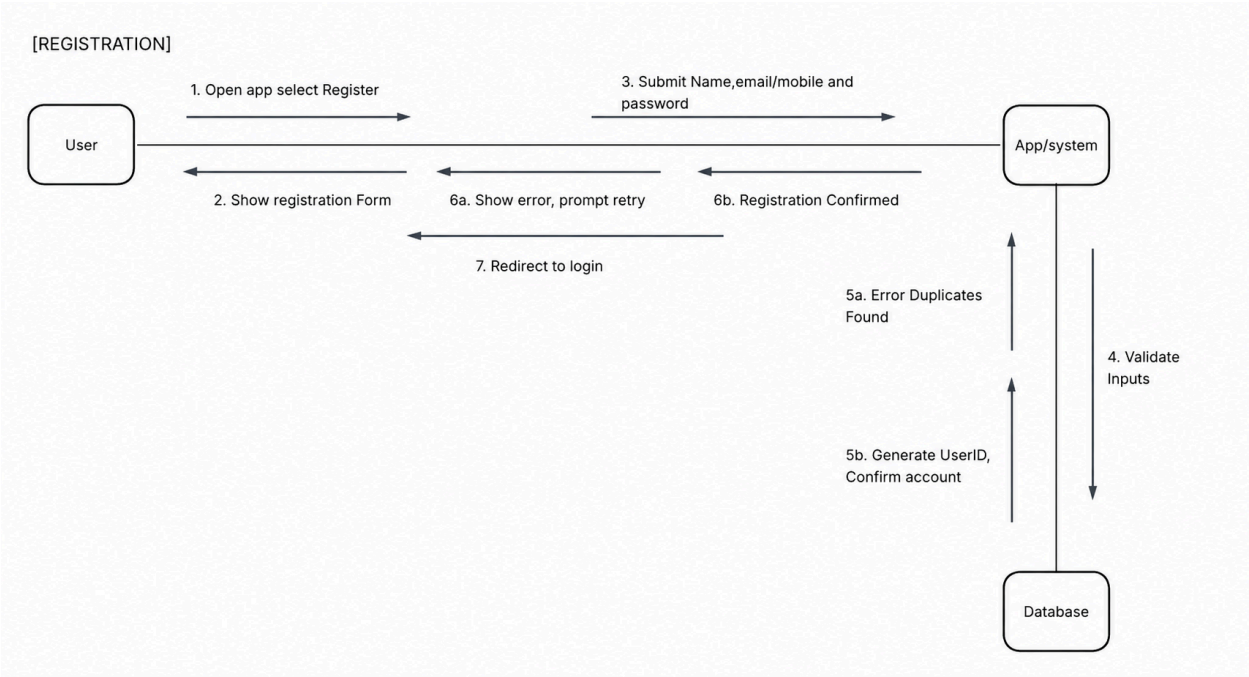
6. Booking, Tenant & Payment Management (Admin)



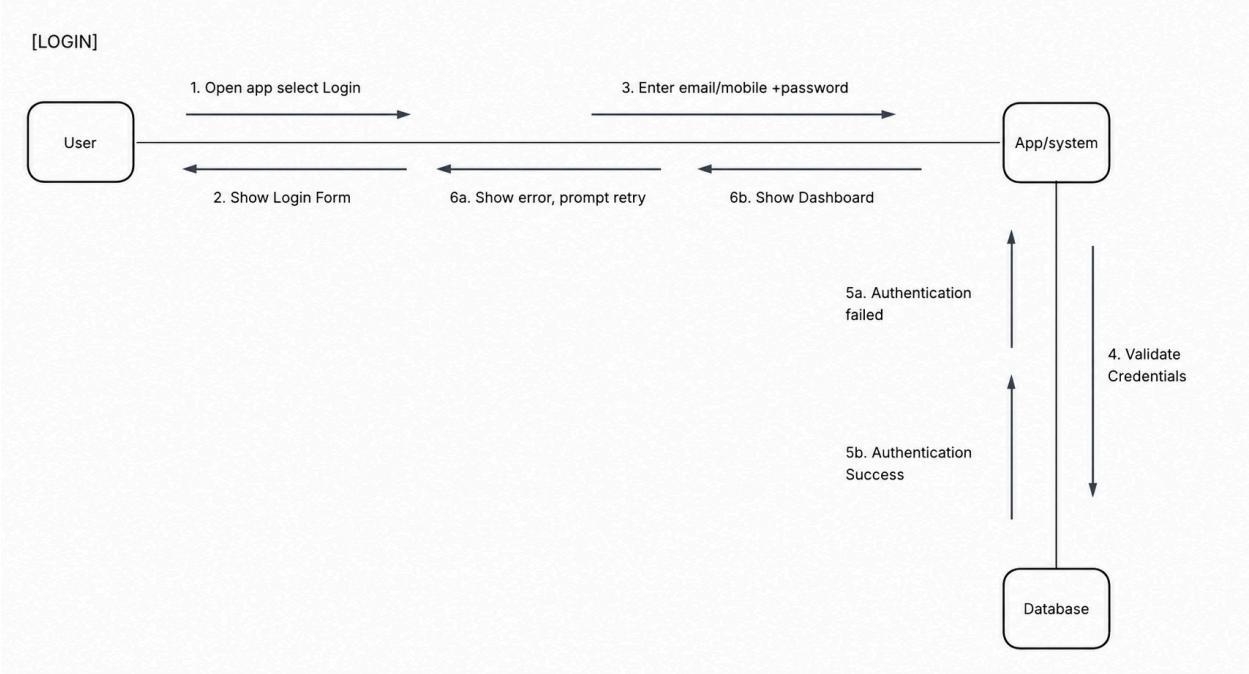
ASSIGNMENT - 06

Collaboration Diagrams

1. User Registration

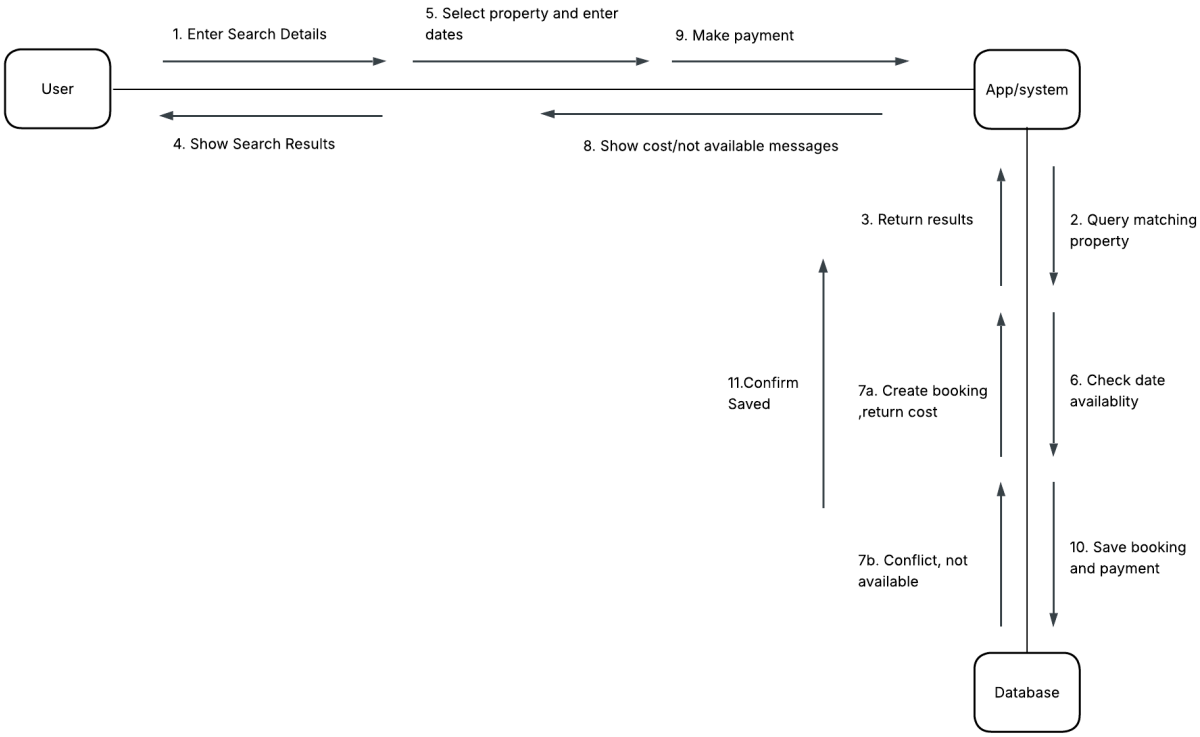


2. User Login



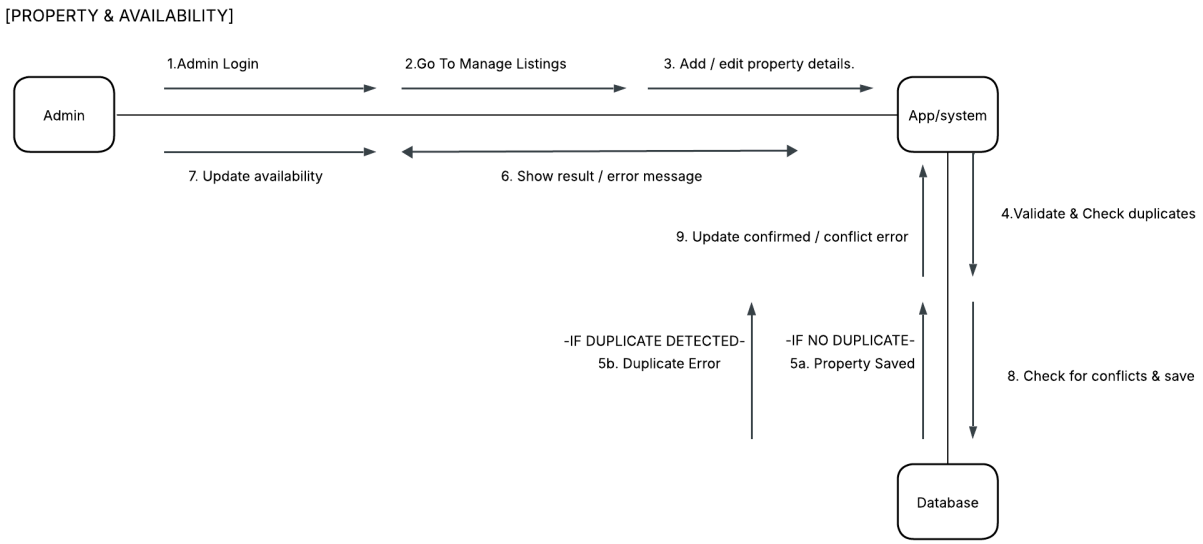
3. Search & Book Property

[SEARCH AND BOOK PROPERTY]

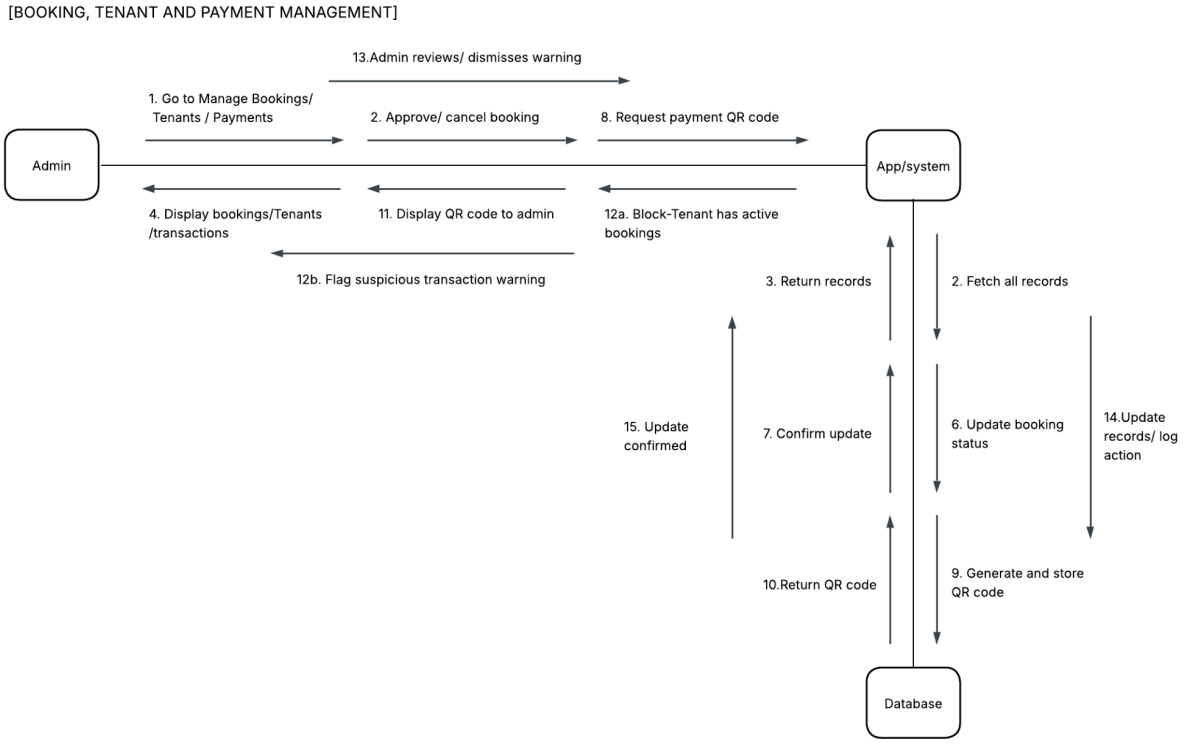


4. View Bookings & Payments

5. Property & Availability Management (Admin)



6. Booking, Tenant & Payment Management (Admin)



ASSIGNMENT - 07

Activity Diagram (Rental Management System)